Niko Weaver

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Education

Relevant Coursework: Space Systems Design, Thermodynamics, Mechatronics, Dynamics

Duke University, Durham NC — Expected Spring 2028

- Bachelor of Science in Engineering
- Major in Mechanical Engineering, Aerospace Certificate, Minor in Math

Waterford School, Sandy UT — August 2020 - June 2024

- High school diploma, Summa Cum Laude.

Technical Skills

Software: Autodesk Fusion CAD, CAM, FEA; Siemens NX CAD; Solidworks CAD; Ansys Fluent CFD; Adobe Photoshop, Illustrator; Microsoft Office suite; Agentic coding;

Languages: MATLAB, Python, Java, Arduino C++.

Projects

Reinforcement Learning with MuJoCo | Duke General Robotics Lab | Summer 2026

- Developed a digital twin my robot in MuJoCo to enable iterative control strategy testing and bridge sim to real gap.
- Trained RL policies to automate and improve locomotion over a heuristic policy

Autonomous UAV | Duke Co-Lab grant project | Summer 2025 – Present

- Engineered an autonomous UAV through three design iterations, leveraging CFD analysis to optimize wing geometry and enhance flight endurance.

RoboSub AUV | Duke Robotics Club | Fall 2024 – Present

- Led design for primary structural components; redesigned the buoyancy system using CAD/FEA simulation, reducing drag by 30% and improving control.
- 7th Place Overall at RoboSub 2025 and 3rd Best Design Report.

Self-Guided Rocket | Independent Project | Summer 2023

- Designed, manufactured, and test-flew a 1-meter canard-guided rocket.
- Implemented a closed-loop control system using IMU sensors and Arduino to maintain stable flight trajectories during high-velocity maneuvers.

Experience

Undergraduate Researcher | Duke General Robotics Lab | Fall 2025 – Present

- Leading the design of an underwater rolling robot, in preparation for publication.
- Developing novel locomotion and communication methods for swarm robotics.

Chief Engineer | Duke Robotics Club | Fall 2025 – Present

- Leading new AUV development for the RoboSub 2027 competition, managing systems integration and performance testing.
- Coordinate cross-functional teams across mechanical, electrical, and software disciplines to ensure cohesive architecture.

Undergraduate Intern | University of Utah FPC LAB | Summer 2025

- Designed a custom 4-axis robot arm for wind tunnel model manipulation.
- Developed MATLAB scripts to simulate real-time operation and automate the collection/analysis of torque and drag data, accelerating testing cycles.